

Measures of centre

Mean (Average)

Median (Middle)

Mode (Most common)

The mean is written as $\bar{x}$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{where } n = \text{number of observations}$$

Heavily affected by anomalies, but easy to calculate

Median not really impacted by anomalies, but more complex calculation

Measures of spread

- range (max - min)
- Standard deviation (= variance)

Average of sum of squared
distances from sample mean

$$\sigma^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- Interquartile range ($Q_3 - Q_1$)